

Software Requirements Specification

Online Video Meetings with Real-Time Translation

Document control	Value
Project team	Void Phantoms
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System scope	Web, Android and Windows access are stated in the current baseline; platform-specific scope requires confirmation
Prepared for	Software Engineering semester project
Approval	Pending project team and instructor review



Document status and reading guide

This is a working requirements specification, not a claim that every requirement has been approved or implemented. It is based on the available developer-team meeting notes, survey analysis, individual interview records and the earlier SRS baseline. The consolidated Requirements Elicitation Report referenced by the baseline was not available among the files inspected for this draft; therefore, full original wording for every FR/NFR/BR/CR identifier could not be independently reproduced.

Label	Meaning	Use in this draft
E — Elicited	Explicitly recorded in available evidence.	Preserve as a baseline requirement or stakeholder need.
I — Inferred	Reasonable implication of an elicited need.	Use cautiously; validate before final approval.
P — Proposed	Draft elaboration to make the system specifiable or testable.	Candidate requirement; team approval required.
O — Open / TBD	The evidence does not settle the decision.	Do not implement as a fixed policy until resolved.

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Glossary

This glossary is placed before the Introduction so that readers can consult project-specific terminology before reading the requirements. Definitions describe intended use in this SRS; unresolved semantics are identified as TBD.

Term	Definition
API	Application Programming Interface.
ASR	Automatic Speech Recognition: conversion of speech to text.
Caption	Text representing spoken words, usually associated with timing and possibly a speaker.
FR	Functional Requirement.
MoM	Minutes of Meeting: a structured summary or record of meeting discussion and decisions.
NFR	Non-Functional Requirement.
OAuth	Authorization framework used by an identity provider; Google OAuth is a stated dependency.
Participant session	A period of participation or connection for a user in a meeting; exact reconnect semantics are TBD.
SRS	Software Requirements Specification.
TBD	To be determined by an explicit project/stakeholder decision.
Translation fallback	Switching to an alternative translation engine or a defined degraded mode when the primary engine fails or is too slow.
LiveKit	The media transport/backend intended for production according to the team discussion.
Data requirement	A requirement defining what data the system stores, relates, validates, retains, exports, restores or deletes.
Event trace	A time-ordered representation of messages/events exchanged between actors or components for a scenario.
ER diagram	Entity–Relationship diagram; a conceptual model of data entities and their relationships.
Interface requirement	A requirement describing how a user or external system interacts with VoidMeet.
Live caption	Text produced from recognized speech during a meeting, normally displayed in time order.
Real-time translation	Conversion of recognized speech or text into a selected target language with a delay suitable for live interaction; measurable limits are TBD.
Security requirement	A requirement that protects identity, meeting access, data, privacy, credentials or system operations.
UML	Unified Modeling Language; a notation for describing software structure and behavior.
Waiting room	A meeting state in which a participant waits for admission by a host or authorized role.
Reconnect	A participant's return after a temporary connection loss; rules for merging attendance intervals are TBD.

Requirement traceability	The ability to link a requirement to its source evidence, design representation and verification evidence.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the intended capabilities, external interfaces, quality expectations, constraints and unresolved decisions for VoidMeet, an online meeting platform intended to support real-time translated communication. It is written for the project team, instructor, developers, testers and reviewers. The specification is intended to provide a common baseline for design, implementation, verification and change control.

The document describes what the system is expected to do and the conditions under which it should operate. It does not claim that the proposed architecture, integrations or performance targets have already been implemented or validated.

1.2 Scope

VoidMeet is intended to let users create and join online meetings, communicate using audio/video and meeting collaboration tools, and access live captions and translated speech/text where supported. The requirements elicitation also identifies meeting access control, participant moderation, screen sharing, chat, reactions, collaborative whiteboard features, meeting outputs, attendance reporting and recovery of relevant meeting state.

The currently stated access targets include web, Android and Windows. Exact platform coverage and feature parity are TBD. The team discussion records LiveKit as the intended production media backend and restricts translation engines to free services under the project constraint. The actual free-service availability, licensing, rate limits and reliability must be verified before implementation decisions are finalized.

1.2.1 In-scope capability areas

- Identity and sign-in, with Google OAuth recorded as a dependency.
- Meeting creation, joining, admission/waiting-room behavior and meeting roles.
- Audio/video participation, participant list, microphone/camera controls and screen sharing.
- Live captions and real-time translation with fallback behavior where a supported engine is available.

- Chat, reactions, collaborative whiteboard, formatted code blocks and screen annotation subject to authorization decisions.
- Recording, transcript, Minutes of Meeting (MoM), output access and retention controls, subject to consent and policy decisions.
- Attendance events, presence duration and authorized host/organizer reports.
- Persistence and restoration of relevant meeting state after a server restart.
- Security controls, rate limiting, protected credentials and operational recovery.

1.2.2 Out of scope or not yet committed

- A guarantee of professional-grade translation accuracy for every language, accent, domain or noisy environment.
- A final supported-language list or guaranteed language-pair coverage.
- A final deployment topology, database choice, storage provider or production hosting budget.
- A committed participant capacity beyond a provisional planning target of approximately 50 concurrent users; this is not a benchmark result.
- A final policy for recording consent, retention duration, deletion, sharing or legal compliance.
- Automated meeting-minute generation as a confirmed feature; manual, automatic and hybrid modes remain open.
- Regional-only screen sharing as a confirmed requirement; survey interest does not by itself define the behavior.

1.3 Document Conventions

The keywords below express the normative force of statements in this SRS. They are used consistently in requirement statements and in this convention table. A requirement statement should identify observable system behavior or a verifiable condition; unresolved values remain explicitly marked TBD.

Keyword	Meaning and usage
SHALL	A mandatory requirement for the specified release once the requirement is approved. A **SHALL** statement is expected to be testable.
SHOULD	A recommended behavior or design objective. Deviations require a recorded rationale and stakeholder review; it is not treated as mandatory unless approved as such.
MAY	An optional capability or permitted approach. It is not mandatory unless another approved requirement makes it so.

TBD	To Be Determined. A value, policy or decision is unresolved. Each TBD **should** have an owner and a closure point before the affected requirement is baselined.
Note	Explanatory or contextual text. A Note is not, by itself, a normative requirement.

1.3.1 Requirement Identifier Conventions

New or normalized requirements should use the identifier families below. The current draft also contains legacy elicitation IDs and provisional elaboration IDs (for example FR-54 and FR-TR-P1). These are retained for traceability until the team approves a mapping; they must not be silently renumbered.

Identifier	Requirement category	Use in VoidMeet
FR-XX	Functional Requirement	A user-visible capability or system behavior, including meeting lifecycle, collaboration, media and outputs.
NFR-XX	Non-Functional Requirement	A measurable quality attribute or operating constraint, such as performance, reliability, usability or maintainability.
IR-XX	Interface Requirement	A requirement for a user interface or an external software, hardware or communication interface.
DR-XX	Data Requirement	A requirement for stored data, data relationships, integrity, retention, export, deletion or recovery.
SR-XX	Security Requirement	A requirement for authentication, authorization, privacy, protection of secrets, auditability or abuse prevention.

1.4 Intended Audiences and Stakeholders

The table identifies the main readers and stakeholders and indicates which parts of the SRS are most relevant to them. Roles are project-level categories; individual approval authority should be confirmed with the course instructor and team.

Audience / stakeholder	Primary interest or responsibility	Most relevant sections
Course instructor / project supervisor	Assess scope, elicitation quality, completeness, feasibility and traceability; resolve academic/project-level decisions.	Document status; Sections 1–3; appendices and review checklist
Project team / requirements owner	Maintain scope, approve requirement wording, prioritize open decisions and preserve links to elicitation evidence.	Document conventions; Sections 1–3; open decisions; traceability
Developers / system designers	Understand required behavior, boundaries, interfaces, data concepts, dependencies and recovery expectations.	Sections 2 and 3; ER/UML models; event traces; constraints
Testers / verification reviewers	Derive test cases from **SHALL** statements, verify observable behavior and record unresolved acceptance thresholds.	Section 3; acceptance evidence; open-decision register

Meeting hosts / organizers	Understand meeting control, admission, moderation, recording/output access and attendance reporting needs.	Scope; user classes; meeting, security, output and attendance requirements
Meeting participants / students / remote users	Understand joining, audio/video, captions, translation, chat, whiteboard and privacy expectations.	Scope; user classes; user-interface and functional requirements
Teachers / lecturers / facilitators	Review collaborative teaching needs, formatted code, annotation, whiteboard export and meeting records.	Functional requirements; collaboration/output requirements; diagrams
Administrator / deployment operator	Understand configuration, secrets, service dependencies, monitoring, deployment and recovery.	Constraints; dependencies; security, reliability and deployment notes
Documentation / support owner	Prepare onboarding, help, troubleshooting and handover materials based on approved behavior.	User documentation; interfaces; support and operational requirements

1.5 Abbreviations and definitions

The project glossary is provided in the front matter before Section 1. It defines the terms and abbreviations used throughout this specification.

1.6 References

Ref.	Reference
SRC-01	VoidMeet Requirements Elicitation Report, updated version, October 2026 — referenced by the baseline but not available in the inspected file set; exact catalogue text must be reattached/verified.
SRC-02	VoidMeet Developer and Project Team Discussion, recorded 21–23 September 2026; revised meeting notes, approval pending.
SRC-03	VoidMeet Survey Data Analysis Report; 10 responses collected around 2–3 October 2026.
SRC-04	Individual interview records: ID-AD-002 (admin Sohail Sarwer), ID-TE-003 (Engr. Sumayyea Salahuddin), ID-TE-004 (Dr. Rehmat Ullah Khattak), ID-ST-001 (student Syed Talha Ali Shah).
SRC-05	VoidMeet SRS Requirements Baseline v0.1, dated 17 October 2026.
SRC-06	VoidMeet developer-team meeting notes in formatted DOCX and revised Markdown versions.

1.7 Outline

Section 2 describes the product context, functions, users, constraints and dependencies. Section 3 specifies interfaces, functional behavior and quality requirements. The appendices contain

conceptual diagrams, open decisions and traceability notes. Diagrams are design-supporting models, not a substitute for the normative requirement statements.

2. Overall Description

2.1 Product context

VoidMeet is a meeting application that coordinates identity, meeting lifecycle, media communication, collaboration tools and meeting outputs. It depends on external identity and media/translation capabilities. The user-facing application and backend must coordinate meeting membership, permissions and persistent outputs. The exact service decomposition and production deployment remain unconfirmed.

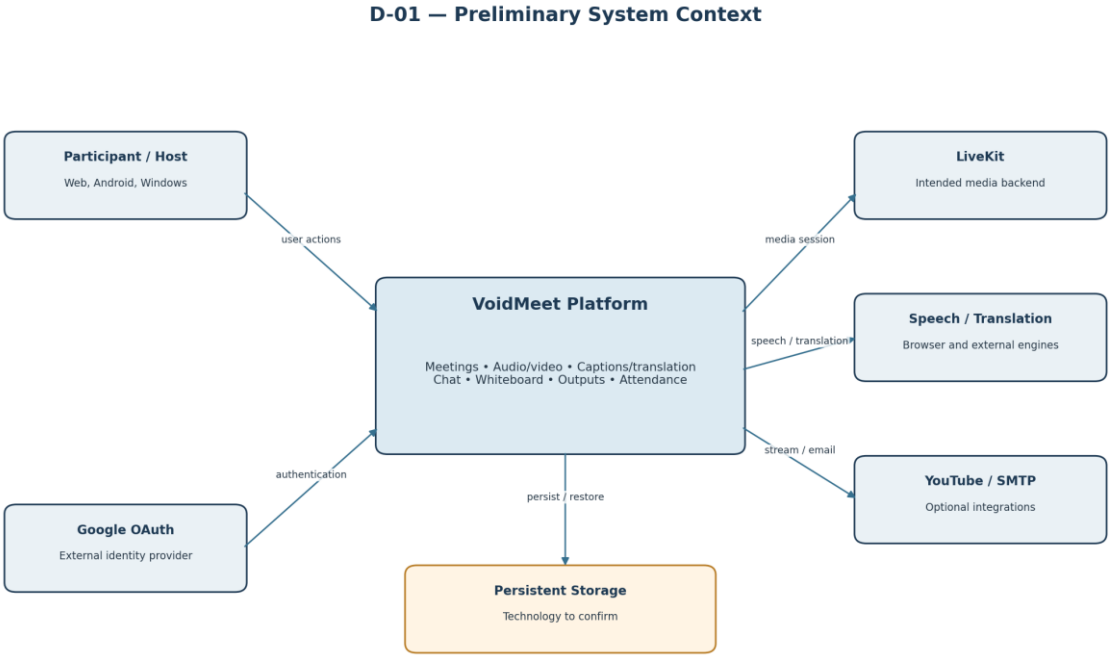


Figure 1. Preliminary system context. Arrows are conceptual; protocols and final deployment are not confirmed.

2.1.1 External actors and services

Actor / service	Purpose	Status / boundary
Meeting participant	Joins a meeting; uses audio/video, chat, captions/translation and collaboration features according to permissions.	Elicited capability family; detailed role permissions require validation.

Host / authorized organizer	Creates or manages a meeting, controls admission and participation, and accesses permitted reports/outputs.	Attendance report restriction is explicit in BR-14.
Administrator	Manages access and system-level risks/policies; monitors relevant users, roles and outputs.	Interview evidence; precise system-admin functions require confirmation.
Google OAuth	Provides identity sign-in.	Dependency recorded in the baseline; OAuth configuration and account policy TBD.
LiveKit	Intended production media transport/backend.	CR-01; implementation choice recorded as intended, not independently validated.
Speech/translation engines	Provide speech recognition and translation capabilities.	Must comply with free-engine constraint; providers, languages, quotas and fallback policy TBD.
Persistence/storage	Stores required meeting state and outputs.	FR-60 requires state restoration; final database/storage design TBD.
Email service	Potentially supports invitations and notifications.	Scheduled-meeting email invitations and SMTP details are open; do not assume enabled.

2.1.2 System boundary

The application boundary includes user-facing meeting functions, meeting/participant coordination, authorization checks, meeting collaboration state, output management and attendance calculations. External identity, media transport, speech recognition, translation, email and durable storage are dependencies or integrations. The precise boundary between application services and third-party services will be finalized during design.

2.2 Product functions

Capability	Description	Trace / evidence
Authentication and identity	Sign in through the agreed identity flow and associate a user identity with meeting actions.	FR-01–03; BR-01; Google OAuth dependency.
Meeting lifecycle	Create or schedule meetings, join via meeting access information, and handle admission.	FR-04–11; scheduled email behavior TBD.
Admission and roles	Waiting room/admission, host controls, participant roles and moderation.	FR-12–18; BR-02–05; admin interview ID-AD-002.
Live communication	Audio/video participation, participant list, screen sharing and controls.	FR-19–27; low-bandwidth and screen-sharing concerns from survey/interviews.

Captions and translation	Produce captions and translated text/audio where available; handle slow or failed engines.	FR-28–33; NFR-02/05/10; BR-10; CR-02.
Chat and reactions	Send meeting messages and use reactions/raise-attention features.	FR-34–40; BR-07/08.
Whiteboard and annotation	Collaboratively draw, use engineering diagrams/code formatting, annotate shared content and export board image.	FR-27; FR-41–45; BR-06; teacher interviews.
Host controls	Manage participants and control meeting capabilities.	FR-46–47; BR-05/09.
Meeting outputs	Create/access recordings, transcripts and MoM under access and retention rules.	FR-48–53; BR-11/13/15; CR-09/10; policies TBD.
Attendance	Record join/leave, calculate duration and present an authorized report.	FR-54–56; NFR-15; BR-14.
Personalization, client apps and support	Preferences, application targets and support reporting.	FR-57–59; BR-12; CR-11; exact details must be verified against SRC-01.
Recovery	Persist and restore meeting settings, whiteboard, chat and relevant participant state after restart.	FR-60; NFR-06; CR-08 is current implementation detail, not an approved architecture decision.

2.3 User classes and characteristics

User class	Goals and expected interaction	Security / permission considerations
Participant / attendee	Join a meeting, communicate, read captions/translation, use chat/reactions and access permitted collaboration features.	May not automatically present, annotate, record or access reports; permission defaults TBD.
Host	Start/manage meeting, admit/remove participants, manage microphones/sharing, and access authorized meeting outputs.	Must be able to respond to uninvited or disruptive joiners; exact controls and host succession TBD.
Authorized organizer / co-host	Assist the host and, where authorized, view attendance or manage outputs.	Role privileges must be explicit; FR-56 and BR-14 restrict attendance report access.
Administrator	Oversee access, roles, system abuse, and the existence/access of outputs.	System-level versus meeting-level administration boundary TBD.
Teacher / presenter	Present material, draw engineering diagrams, annotate shared content and retain board material after class.	Annotation and board clearing/export permissions TBD.
Student / learner	Follow spoken content through captions/translation, use chat/board, and access permitted learning outputs.	Recording/transcript visibility and retention policy TBD.

Guest / unauthenticated user	Potentially join through a meeting link.	Guest access is not confirmed; protected actions require an agreed identity/access policy.

2.3.1 User characteristics

The survey and interviews suggest a mix of university students, teachers, general users and remote-work participants. The interface should support users with varied technical confidence, users on weak connections, and users who need captions or translation. These are qualitative design signals, not a validated accessibility conformance claim.

2.4 Operating environment

- Target access environments currently include web, Android and Windows; supported versions, browsers and device requirements are TBD.
- Real-time audio/video depends on network connectivity, device permissions, available camera/microphone hardware and the media transport service.
- Speech recognition and translation depend on external engine availability, language support, quota/rate limits and network latency.
- Persistent meeting state and meeting outputs require a storage mechanism; database and object-storage choices are not confirmed.
- The intended production media backend is LiveKit (CR-01). The deployment diagram is deferred until the hosting topology is selected.

2.5 Design and implementation constraints

Constraint ID	Constraint	Interpretation / validation needed
CR-01	Production media transport uses LiveKit.	Confirm SDK, hosting mode, credentials and deployment architecture.
CR-02	Only free translation engines are permitted under the current project constraint.	Free tier quotas, terms, service availability and language support must be verified; “free” does not imply unlimited or guaranteed uptime.
CR-06	API keys and credentials must remain outside source code.	Use environment/secrets configuration; no credentials in committed source or logs.
CR-08	Current meeting state is held in server memory with file persistence.	This is recorded as current implementation detail; no architecture change was decided. Validate whether it can satisfy FR-60/NFR-06.

CR-05/07 and CR-09/10/11	These IDs appear in the baseline's constraint family.	Exact wording was not available in inspected sources; retrieve SRC-01 before final approval rather than guessing.

2.6 User documentation

Proposed documentation deliverables for the semester project are: a short getting-started guide, a meeting-host guide covering admission/roles/recording and output access, a participant guide covering audio/video/captions/translation/chat/whiteboard, and an administrator/deployment guide for configuration and secrets. These deliverables are proposed and should be confirmed against project marking requirements.

2.7 Assumptions and dependencies

- Google OAuth is available and configured for the agreed client platforms (dependency recorded in the baseline).
- LiveKit can be deployed or accessed within the project's available resources.
- At least one free speech-recognition/translation route is available for the languages selected by the team; language list remains TBD.
- Users grant browser/device permissions for microphone, camera and screen capture as required.
- Network quality can vary; the system should provide understandable degraded/failure feedback rather than imply guaranteed service.
- Survey findings are treated as early discovery evidence from ten respondents, not as population-wide statistics.
- The exact wording of FR-01–53 and NFR-01–14 should be checked against the original consolidated elicitation report before this draft is approved.

3. Specific Requirements

Normative convention: the keywords SHALL, SHOULD and MAY are defined in Section 1.3. Requirements labelled P are proposed elaborations and require review; source-supported requirements are labelled E where indicated. Acceptance criteria depending on unresolved architecture or policy are explicitly TBD.

3.1 External interface requirements

3.1.1 User interfaces

- UI-01 (P): The application **shall** provide a meeting view that makes the participant list, current meeting status and available meeting controls discoverable without requiring users to navigate away from the meeting.
- UI-02 (P): The application **shall** visually distinguish host/organizer controls from ordinary participant controls according to the final role-permission matrix.
- UI-03 (P): The application **shall** provide visible feedback when a participant is waiting for admission, is denied entry, loses connection, reconnects, or cannot use a requested capability.
- UI-04 (P): When captions or translation are enabled, the interface **shall** display the selected source/target language and present caption/translation status or failure feedback.
- UI-05 (P): When a meeting is being recorded or an output is being generated, the interface **shall** display the agreed notification/consent state. Exact consent flow is TBD (OD-04).
- UI-06 (P): The whiteboard interface **shall** provide drawing and selection controls and, if approved, controls for formatted code blocks, screen annotation and image export.
- UI-07 (P): The attendance report interface **shall** show participant name, join time, leave time and presence duration for authorized roles (FR-56). Export buttons **shall** not be shown unless export is approved (OD-11).
- UI-08 (P): Error messages **shall** describe the user-visible problem and, where practical, a recovery action without exposing credentials, private provider responses or internal stack traces.
- UI-09 (P): The interface **shall** adapt to supported screen sizes on the approved web/mobile/desktop targets; exact minimum viewport sizes and accessibility criteria are TBD.

3.1.2 Software interfaces

Interface	Requirement / boundary	Open specification
Identity provider	The system **shall** use the approved Google OAuth flow for sign-in if this dependency remains in scope.	Client IDs, redirect URIs, account restrictions, logout/revocation behavior TBD.
Media transport	The production media path **shall** use LiveKit as intended by CR-01.	SDK/API version, self-hosted vs managed service, tokens and room mapping TBD.
Speech recognition	The system **may** integrate a speech-to-text engine to generate captions.	Provider, supported browsers, languages, confidence handling and limits TBD.

Translation	The system **shall** use only translation engines allowed by CR-02 and support fallback behavior consistent with NFR-05.	Primary/fallback providers, language pairs, timeout boundary and failover policy TBD.
Persistence	The system **shall** persist the meeting state required by FR-60 and restore it after restart.	Database, schema, file/object storage, transaction and backup strategy TBD.
Email	If scheduled email invitations are approved, the system **shall** use a configured mail service.	SMTP/provider and delivery/retry policy TBD; invitation feature itself remains open.
Client operating systems	The baseline lists web, Android and Windows access.	Version support, packaging and feature parity require confirmation.

3.1.3 Communications interfaces

- NET-01 (P): Application control traffic **shall** use encrypted transport in production. Exact TLS version and deployment configuration are design decisions to verify.
- NET-02 (P): Real-time media **shall** use the selected LiveKit transport configuration and **shall** report a meaningful failure state when connection establishment fails.
- NET-03 (P): API requests **shall** be authenticated/authorized where the operation is protected; the server **shall** not rely solely on client-side hiding of controls.
- NET-04 (E/P): Rate limiting is required (NFR-04). The limits, identity/IP scope, burst policy and user-facing response must be specified and tested.

3.1.4 Hardware interfaces

The application interfaces with supported microphones, cameras, speakers/headphones, and screen-capture facilities through the host operating system or browser APIs. Exact device compatibility is platform-dependent and shall be documented after target browsers and operating-system versions are selected. No custom hardware interface is established by the available evidence.

3.2 Functional requirements

The requirement-family ranges below follow the IDs recorded in the available baseline. For many ranges, the exact original requirement text is not present in the available source files; the detailed statements in this section are proposed elaborations for review, not a replacement for the missing

source catalogue. FR-54, FR-55, FR-56 and FR-60 have explicit wording in the developer-team meeting notes and are treated as elicited requirements.

3.2.1 Authentication and identity (FR-01–FR-03)

- FR-AUTH-P1 (P; trace to FR-01–03): The system **shall** provide the approved sign-in flow and associate authenticated actions with the corresponding user identity.
- FR-AUTH-P2 (P): The system **shall** reject protected operations when the caller is unauthenticated or the identity session is invalid.
- FR-AUTH-P3 (P): The system **shall** not expose OAuth client secrets, API keys or other credentials in client-visible source bundles or logs. CR-06 explicitly requires credentials to remain outside source code.
- TBD: guest access, account lifecycle, profile fields, account deletion and session expiration policy.

3.2.2 Meeting creation, scheduling and joining (FR-04–FR-11)

- FR-MTG-P1 (P; trace to FR-04–11): An authorized user **shall** be able to create a meeting and receive the meeting identifier or join information needed to invite participants.
- FR-MTG-P2 (P): The system **shall** validate meeting access information and provide a clear result when a meeting is invalid, unavailable or closed.
- FR-MTG-P3 (P): If waiting-room admission is enabled, a joining participant **shall** remain pending until an authorized host admits or rejects the participant.
- FR-MTG-P4 (P): The system **shall** prevent a participant who has been denied or removed from immediately bypassing that decision using the same active admission state. Re-entry policy is TBD.
- TBD: scheduled meeting fields, calendar integration, email invitations, recurring meetings, guest joining, meeting expiry and late joining.

3.2.3 Roles, admission and moderation (FR-12–FR-18)

- FR-ROLE-P1 (P; trace to FR-12–18): The system **shall** apply meeting permissions on the server for admission, participant removal, screen sharing, annotation, recording and access to meeting outputs.
- FR-ROLE-P2 (P): The host **shall** be able to view the current participant list and remove a participant when authorized. This is supported by the administrator interview, which describes disruptive/uninvited participants as a concrete problem.

- FR-ROLE-P3 (P): The interface **shall** communicate participant roles and permission-relevant status in a way that reduces ambiguity for hosts and participants.
- FR-ROLE-P4 (P): The system **shall** define the behavior if a host disconnects or leaves while the meeting continues. Host succession rules are TBD.
- TBD: exact role names, whether co-hosts exist, permission defaults, participant self-mute controls, and whether hosts can delegate individual permissions.

3.2.4 Audio, video and screen sharing (FR-19–FR-27)

- FR-MEDIA-P1 (P; trace to FR-19–27): A participant with permission and available hardware **shall** be able to enable or disable their microphone and camera during a meeting.
- FR-MEDIA-P2 (P): The system **shall** indicate whether local microphone/camera capture is enabled and communicate permission or device failures.
- FR-MEDIA-P3 (P): An authorized participant **shall** be able to start and stop screen sharing; other participants **shall** receive the shared content while the share is active.
- FR-MEDIA-P4 (P): The host or authorized role **shall** be able to restrict screen sharing according to the approved permission model.
- FR-MEDIA-P5 (P): The system **should** remain usable under constrained network conditions and provide feedback when audio/video quality is degraded (survey signal; measurable targets TBD).
- TBD: selective-window/region sharing, simultaneous sharers, maximum resolution/frame rate, mobile screen-sharing limitations and platform parity. Survey interest in region-based sharing is not yet a confirmed requirement.

3.2.5 Captions and real-time translation (FR-28–FR-33)

- FR-TR-P1 (P; trace to FR-28–33): When captioning is enabled and a supported speech-recognition engine is available, the system **shall** present recognized speech as time-ordered caption segments.
- FR-TR-P2 (P): When a supported target language is selected, the system **shall** attempt to translate recognized caption text into that target language.
- FR-TR-P3 (P): The system **shall** communicate when captioning or translation is unavailable, delayed, rate-limited or has failed; it **shall** not silently present stale text as current.
- FR-TR-P4 (E/P; NFR-05): If the primary translation engine fails or becomes too slow, the system **shall** attempt the configured fallback path or enter a defined degraded state.
- FR-TR-P5 (P): The system **shall** restrict engine use to services permitted by CR-02 and BR-10.

- FR-TR-P6 (P): Speech input size/rate **shall** be bounded to reduce overload and abuse as required by NFR-04.
- TBD: supported source/target languages, whether translated audio is required, speaker labels, transcript persistence, translation confidence, the meaning of the approximately eight-second timeout in NFR-02, and objective accuracy/latency test methodology.

3.2.6 Chat and reactions (FR-34–FR-40)

- FR-COM-P1 (P; trace to FR-34–40): A meeting participant **shall** be able to send a text message to the meeting chat when permitted.
- FR-COM-P2 (P): The system **shall** associate each persisted chat message with its author and meeting context where persistence is enabled.
- FR-COM-P3 (P): The system **shall** provide the agreed reaction/attention mechanism and update the meeting view for connected participants.
- FR-COM-P4 (P): The system **shall** enforce chat permissions and rate limits to mitigate abuse.
- TBD: direct/private messages, edit/delete behavior, message history for late joiners, attachment support, retention and moderation rules.

3.2.7 Whiteboard and screen annotation (FR-41–FR-45; related FR-27)

- FR-WB-P1 (P; trace to FR-41–45): Authorized participants **shall** be able to add drawing objects to a shared meeting whiteboard and see updates from other authorized participants.
- FR-WB-P2 (P): The system **shall** preserve whiteboard content as part of the relevant meeting state and make it available to authorized users according to the retention policy.
- FR-WB-P3 (E/P): The whiteboard **shall** support export of its visible content as an image (FR-45), reflecting the teacher interview request.
- FR-WB-P4 (P): The system **shall** support formatted code blocks on the whiteboard if the team confirms that capability; supported programming languages are TBD (Q-5).
- FR-WB-P5 (P): If screen annotation is enabled, the system **shall** synchronize annotations with the shared screen and restrict annotation to approved roles (Q-4).
- FR-WB-P6 (P): The system **shall** define behavior for concurrent edits, clearing the board, late joiners and recovery after restart.
- TBD: drawing tool set, object deletion/undo, board versioning, export formats/resolution, supported code languages and annotation permission defaults.

3.2.8 Host controls (FR-46–FR-47)

- FR-HOST-P1 (P; trace to FR-46–47): The host **shall** be able to use the approved participant-management controls during an active meeting.
- FR-HOST-P2 (P): The system **shall** enforce the host's changes to meeting permissions for subsequent actions and provide visible confirmation or failure feedback.
- TBD: exact controls, co-host delegation, host transfer, room lock and behavior when the host becomes disconnected.

3.2.9 Recording, transcripts, minutes and output access (FR-48–FR-53)

- FR-OUT-P1 (P; trace to FR-48–53): If recording is included in the approved release scope, the system **shall** provide a host-authorized way to start and stop a meeting recording.
- FR-OUT-P2 (P): The system **shall** associate each generated output with the meeting and enforce access rules before displaying or downloading it.
- FR-OUT-P3 (P): The system **shall** provide transcripts when transcript generation is enabled and a supported speech-recognition path is available.
- FR-OUT-P4 (P): The system **shall** support Minutes of Meeting as an output only after the team confirms whether the content is manually entered, automatically generated or hybrid (Q-3).
- FR-OUT-P5 (P): The system **shall** not make meeting outputs public by default; output access must follow an approved authorization policy.
- FR-OUT-P6 (P): Before recording or transcription begins, the system **shall** follow the approved notification/consent policy. This is not fully specified in the evidence and must be resolved (OD-04).
- TBD: consent capture, recording indicator, output download/share permissions, retention and deletion, storage limits, transcript speaker attribution, MoM generation and streaming semantics.

3.2.10 Attendance and reporting (FR-54–FR-56)

- FR-54 (E): The system **shall** record each participant's join time and leave time.
- FR-55 (E): The system **shall** calculate each participant's total presence duration.
- FR-56 (E): The host or authorized organizer **shall** be able to view an attendance report containing participant name, join time, leave time and presence duration.
- ATT-P1 (P): The system **shall** record enough join, leave and reconnect event information to avoid double-counting a participant's presence when a transient reconnection occurs. Exact grace-period and interval-merging rules are TBD.

- ATT-P2 (E): Attendance events **shall** be recorded consistently enough to calculate reliable presence duration (NFR-15).
- ATT-P3 (E): Attendance reports **shall** be restricted to authorized meeting roles (BR-14).
- TBD: timezone/display convention, report export formats, how missing leave events are closed, whether simultaneous devices count as one or multiple attendances, and the reconnect grace period.

3.2.11 Personalization, client applications and support (FR-57–FR-59)

- FR-APP-P1 (P; trace to FR-57–59): The system **shall** expose only the personalization and support functions confirmed by the original requirement catalogue and approved scope.
- FR-APP-P2 (P): A support request, if in scope, **shall** validate required fields and provide confirmation that the request was submitted or a clear failure message.
- TBD: exact user preferences, supported application packaging, support form fields, SMTP configuration and expected response workflow. Exact original text for these IDs must be recovered from SRC-01.

3.2.12 Persistence and recovery (FR-60)

- FR-60 (E): The system **shall** persist and restore meeting settings, whiteboard content, chat and relevant meeting state after a server restart.
- REC-P1 (P): Following restart, the system **shall** restore persisted data without exposing it to users who are not authorized for the meeting.
- REC-P2 (P): If some state cannot be restored, the system **shall** report the degraded state and avoid falsely indicating that missing data has been recovered.
- REC-P3 (P): Recovery tests **shall** include chat, whiteboard, settings and relevant participant information as identified in the team discussion.
- TBD: maximum recovery time, recovery-point objective, treatment of live media streams, in-flight messages, recordings, external engine sessions and consistency conflicts. The current in-memory/file persistence detail (CR-08) is not a final architecture decision.

3.3 Non-functional requirements

Where source identifiers and exact thresholds are unavailable, proposed requirements are labelled P and must be confirmed. The original catalogue should be restored before final submission so the wording and IDs of NFR-01–NFR-15 can be checked against this elaboration.

ID	Requirement / current interpretation	Status and acceptance evidence

NFR-01	Usability: the system **should** provide an understandable meeting interface for participants with varied technical experience.	Baseline family; proposed elaboration. Usability test tasks and success threshold TBD.
NFR-02	Translation latency: baseline notes approximately eight seconds, but the start/end measurement points are unclear.	Elicited threshold fragment; define measurement boundary, percentile and test conditions before acceptance.
NFR-03	Exact original wording not available in inspected sources.	Retrieve SRC-01; do not infer.
NFR-04	Rate limits and bounded speech input **shall** reduce overload and abuse.	Elicited in team notes. Test limit behavior, response code/message and resource protection; numeric limits TBD.
NFR-05	Translation engines **shall** support fallback behavior.	Elicited in team notes by reference to SRC-01. Test primary timeout/failure and fallback/degraded state.
NFR-06	The application **shall** restart after crashes and restore persisted meeting state.	Elicited in team notes. Test restart and verification of settings, whiteboard, chat and relevant participant state; recovery time TBD.
NFR-07	Low-bandwidth usability is a quality concern identified by survey.	Exact source text/threshold unavailable; test conditions and degraded-mode requirements TBD.
NFR-08	Exact original wording not available in inspected sources.	Retrieve SRC-01; do not infer.
NFR-09	Privacy is a quality concern for recording, transcripts and meeting outputs.	Exact original wording not available; define threat model, access rules and retention policy.
NFR-10	Caption/translation quality or operation is a baseline quality concern.	Exact original wording not available; define supported languages, accuracy/latency method and failure criteria.
NFR-11–NFR-13	Exact original wording not available in inspected sources.	Retrieve SRC-01; do not infer.
NFR-14	Usability/low-bandwidth concern appears in baseline family.	Exact original wording unavailable; verify source and measurable acceptance criteria.
NFR-15	Join/leave events **shall** be recorded consistently enough to calculate reliable presence duration.	Elicited in team notes. Test normal join/leave, transient reconnect, duplicate events, abnormal disconnect and restart recovery.

3.3.1 Security and privacy

- SEC-P1 (P): Protected operations **shall** be authorized on the server using the user's identity, meeting membership and applicable role/permission state.
- SEC-P2 (E/P): API keys and credentials **shall** remain outside committed source code (CR-06); secrets **shall** not be returned to clients or written to ordinary logs.
- SEC-P3 (P): The system **shall** apply rate limiting to abuse-sensitive endpoints and bounded speech input as required by NFR-04.
- SEC-P4 (P): Attendance reports and meeting outputs **shall** be accessible only to roles authorized by the approved policy (BR-14 and interview evidence).
- SEC-P5 (P): Recording/transcription notification, consent, retention and deletion rules **shall** be approved before production use; policy remains TBD.
- SEC-P6 (P): Logs **should** avoid storing unnecessary meeting content, access tokens or secret values. Log retention and audit requirements TBD.

3.3.2 Reliability and recovery

- REL-P1 (E/P): Translation processing **shall** provide fallback behavior when an engine fails or exceeds the configured timeout (NFR-05).
- REL-P2 (E): The application **shall** restart after crashes and restore persisted meeting state (NFR-06).
- REL-P3 (E): Attendance tracking **shall** handle join/leave/reconnect events consistently enough to calculate presence duration (NFR-15).
- REL-P4 (P): Recovery behavior **shall** be tested with persisted chat, whiteboard, meeting settings and relevant participant information.
- TBD: uptime target, recovery-time target, recovery-point target, backup schedule, disaster-recovery scope and acceptable loss of transient media state.

3.3.3 Performance and capacity

- PERF-P1 (P): The team **shall** define and test a target concurrent-meeting/participant workload before claiming capacity. Approximately 50 concurrent users is a planning estimate only, not a validated capacity guarantee.
- PERF-P2 (P): The team **shall** measure join latency, media start time, caption delay, translation delay and server resource use under documented network and workload conditions.

- PERF-P3 (P): Rate limits and bounded speech payloads **shall** prevent unbounded processing work.
- TBD: exact p50/p95/p99 latency, target concurrency, test hardware, network profile and acceptable error rate.

3.3.4 Compatibility, accessibility and maintainability

- COMP-P1 (P): The team **shall** publish a supported browser/OS/device matrix before release.
- ACC-P1 (P): Controls **shall** have clear labels and keyboard focus behavior on supported web targets; formal accessibility conformance level is TBD.
- MAINT-P1 (P): Provider credentials and environment-specific configuration **shall** be externalized from source code (CR-06).
- MAINT-P2 (P): Integrations with speech/translation engines **shall** be isolated sufficiently to permit configuration or replacement without rewriting unrelated meeting features.
- MAINT-P3 (P): The system **shall** document required environment variables, external services, deployment steps and recovery procedures before final handover.

Appendix A — Data and diagram models

The following models are conceptual proposals to support requirements analysis. They are not a finalized database schema, implementation class diagram or deployment architecture. Cardinalities, field constraints and storage technology must be confirmed during design.

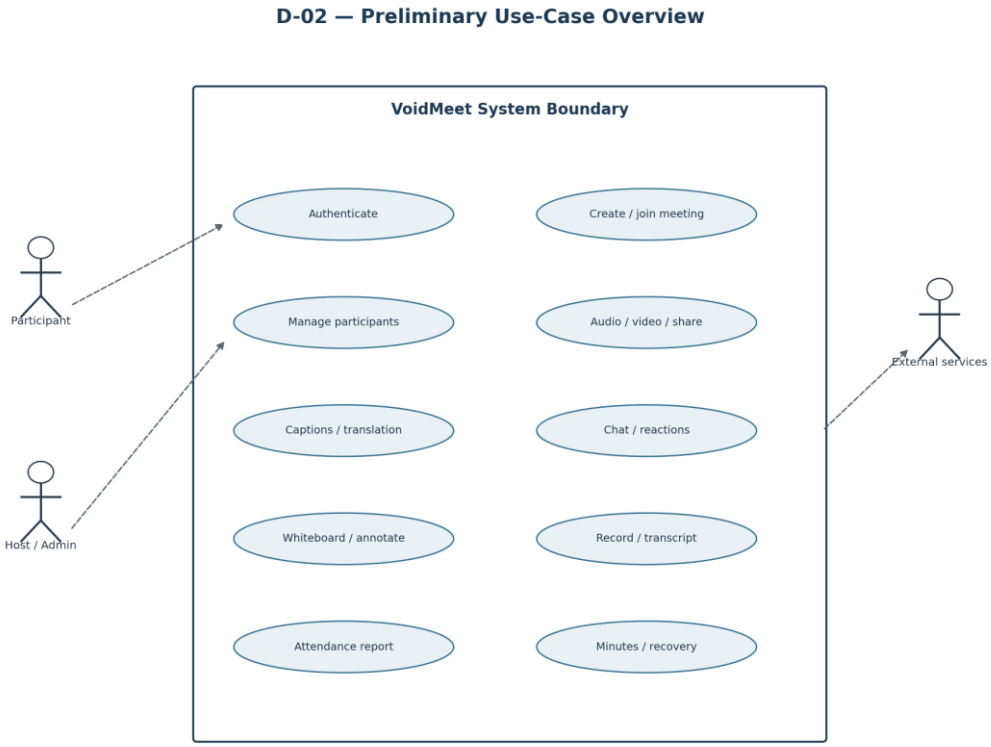


Figure A-1. Preliminary use-case overview.

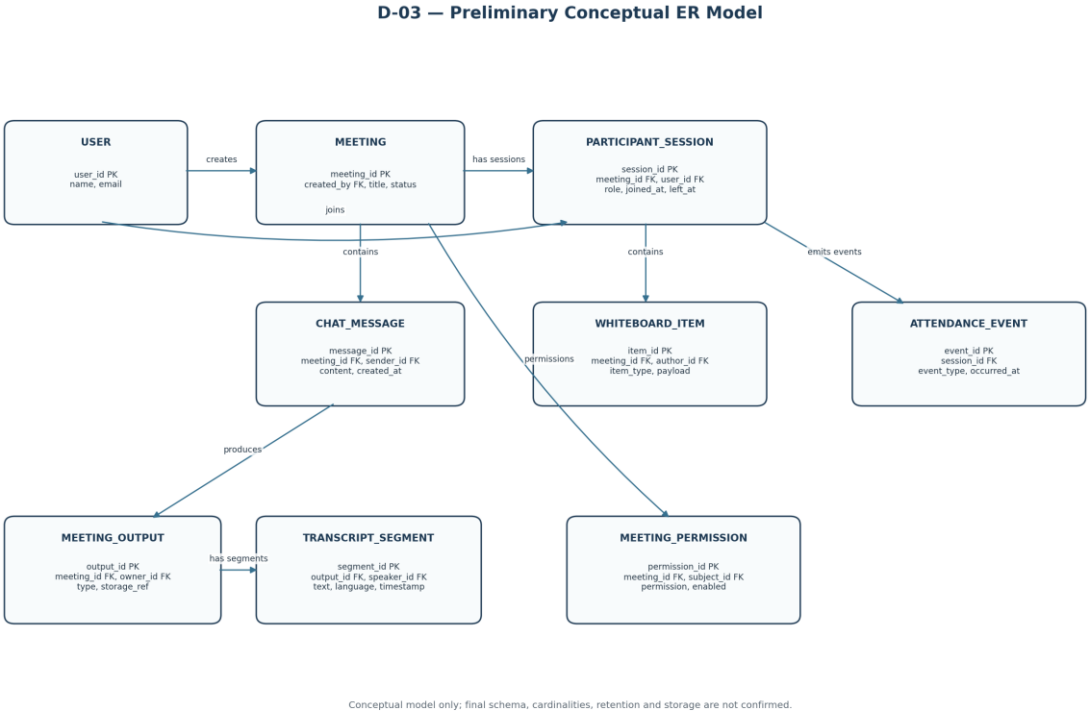


Figure A-2. Conceptual ER model; entity relationships and persistence are proposed.

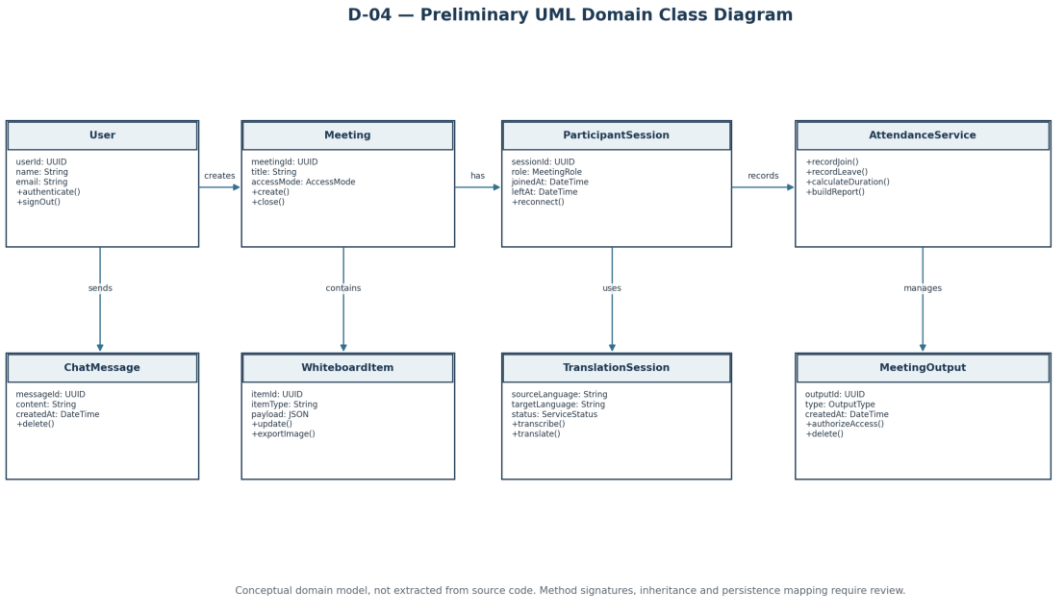


Figure A-3. Conceptual UML domain class diagram; not reverse-engineered from code.

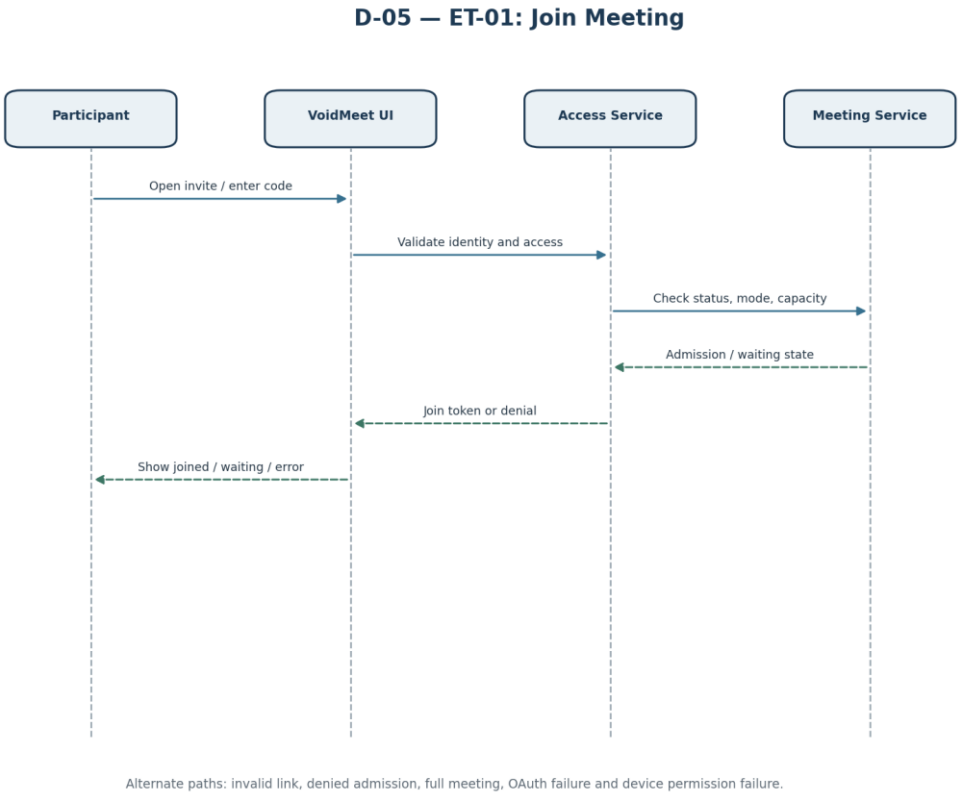


Figure A-4. Event trace: joining a meeting.

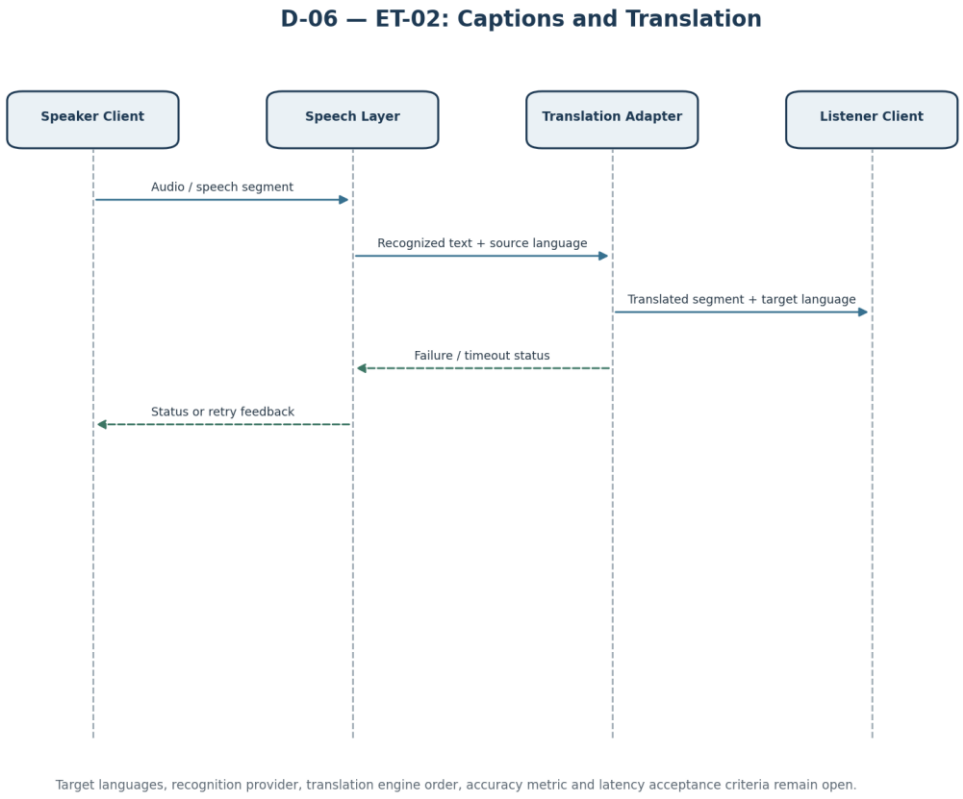


Figure A-5. Event trace: captions and translation, including failure handling.

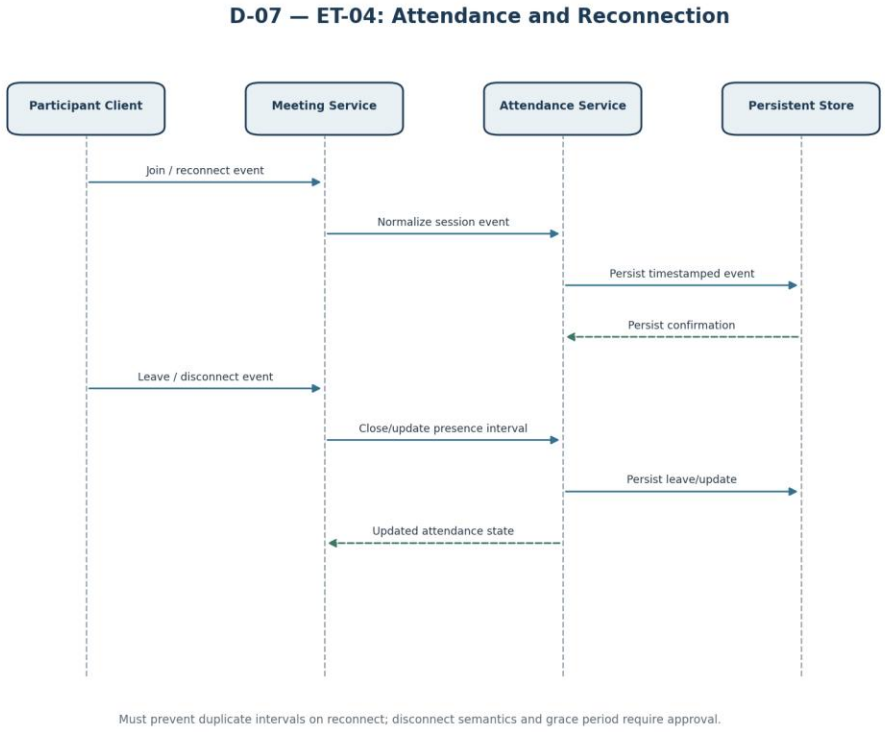


Figure A-6. Event trace: attendance and reconnect handling.

A.1 Conceptual data entities

Entity	Purpose	Fields/relationships to decide
USER	Authenticated identity/profile.	Provider subject, display name, profile fields, account lifecycle and guest identity.
MEETING	Meeting room or scheduled meeting.	Owner, state, creation/schedule fields, admission settings, lifecycle and output policy.
PARTICIPANT_SESSION	A participant's connection/participation session.	Join/leave times, reconnect linkage, device/session ID, termination reason.
ATTENDANCE_EVENT	Timestamped join/leave/reconnect event.	Event type, timestamp source, deduplication key and event origin.
CHAT_MESSAGE	Meeting message.	Author, timestamp, body, edit/delete state, persistence and retention.
WHITEBOARD_ITEM	Collaborative drawing or formatted object.	Geometry/content, author, sequence/version, deletion and export representation.
MEETING_OUTPUT	Recording, transcript, minutes or report.	Output type, creator, storage reference, access policy, consent state, retention/deletion.
TRANSCRIPT_SEGMENT	Timestamped recognized or translated text.	Source/target language, speaker attribution, confidence, timestamps and variants.
MEETING_PERMISSION	Meeting/participant-level permission.	Role, capability, scope, grantor, defaults and override precedence.

A.2 Event trace coverage

Trace	Main path	Alternatives and exception cases to test
ET-01 Join meeting	Validate join request → authenticate/resolve identity → check meeting/access → waiting room or admit → establish participant/media session → record attendance join.	Invalid link, expired/closed meeting, denied admission, missing identity, media failure, duplicate join request, participant reconnect.
ET-02 Captions/translation	Receive audio → speech recognition → caption segment → optional translation → display supported output.	Engine timeout, provider rate limit, unsupported language, malformed/empty result, fallback unavailable, late/stale result.
ET-04 Attendance	Record join/leave events → deduplicate/reconcile reconnects → calculate duration → authorize report access.	Missing leave event, duplicate events, reconnect during grace period, multiple devices, restart, unauthorized report request.

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A.3 Further diagrams recommended

- Activity diagram for admission and participant removal, including waiting-room and denied-access branches.
- Activity diagram for recording/transcription consent, start/stop, output generation and authorized access.
- Activity diagram for translation engine failure/fallback and degraded-mode feedback.
- Deployment diagram only after the actual hosting topology, database/storage and media deployment are selected.

Appendix B — Open decisions and validation plan

ID	Decision needed	Why it matters	Suggested owner/status
OD-01	Supported source/target languages.	Defines speech/translation integration, UI and tests.	Project team/technical lead — open.
OD-02	Meaning and acceptance threshold of approximately 8-second translation timeout.	Needed for testable latency requirement.	Technical lead — open.
OD-03	Target meeting capacity and benchmark conditions.	Prevents unsupported capacity claims.	Technical lead — open.
OD-04	Recording/streaming/transcription notification and consent.	Privacy, trust and legal/ethical expectations.	Team/supervisor — open.
OD-05	Retention/deletion rules for attendance, chat, transcripts, recordings and minutes.	Storage and access control.	Team/supervisor — open.
OD-06	Manual, automatic or hybrid Minutes of Meeting.	Determines generation workflow and quality responsibility.	Project team — open.
OD-07	Screen annotation roles and supported code languages.	Controls misuse and determines whiteboard implementation.	Project team/instructor — open.
OD-08	Reconnect grace period and attendance interval rules.	Avoids duplicate or inaccurate presence duration.	Technical lead — open.
OD-09	Database, file/object storage and recovery architecture.	Required for FR-60 and NFR-06.	Technical lead — open.
OD-10	Scheduled meeting email invitations.	Defines email dependency and scheduling scope.	Project team — open.
OD-11	Attendance report export (view only, CSV, PDF or other).	Affects UI and data export controls.	Project team — open.
OD-12	Deployment topology and production hosting.	Needed for deployment diagram and operations.	Technical lead — open.
OD-13	Guest access and protected-action policy.	Identity and admission security.	Project team — open.
OD-14	Region/window-only screen sharing.	Survey interest does not specify exact feature behavior.	Technical lead — open.
OD-15	Exact supported platforms and version matrix.	Required for compatibility and test planning.	Project team — open.
OD-16	Exact source wording for FR-01–53, NFR-01–14, BR-01–15 and CR-01–11.	Avoids losing original requirements or misattributing proposed elaborations.	Document owner — blocked until SRC-01 is available.

B.1 Recommended validation sequence

1. Recover and verify the original consolidated requirements elicitation report; compare every original ID and wording with this draft.
2. Hold a short team review to confirm scope, role names, platform targets and the meaning of "real-time translation".
3. Resolve OD-04 and OD-05 before implementing recording/transcript/output access.
4. Resolve OD-01, OD-02 and provider quotas before claiming translation coverage or latency.
5. Resolve OD-08 and OD-09 before accepting attendance accuracy and crash-recovery behavior.
6. Define measurable NFR acceptance criteria and create a test case for every shall-statement.
7. Update ER/UML/event-trace models when requirements or architecture decisions change; keep traceability synchronized.
8. Obtain project-team and instructor review/approval and record the version/date of the approved baseline.

Appendix C — Traceability and source limitations

Evidence source	Supported finding	Requirement families / IDs	Caveat
Survey (10 responses)	Interest in live captions, translation, whiteboard, recording, privacy and low-bandwidth support.	FR-28–33; FR-41–45; FR-48–53; NFR quality families.	Small, student-heavy sample; signals interest, not population-level demand.
ID-TE-004: Dr. Rehmat Ullah Khattak	Code formatting and annotation over screen share; need to manage access/roles.	FR-27, FR-44; role/moderation families.	Interview notes are written; no audio/video recording.
ID-TE-003: Engr. Sumayyea Salahuddin	Whiteboard usability for diagrams and export board as image.	FR-41–45, especially FR-45.	Interview notes are written; no audio/video recording.
ID-ST-001: Syed Talha Ali Shah	Transcripts and Minutes of Meeting.	FR-50–52; BR-15.	Generation mode and retention remain open.
ID-AD-002: Sohail Sarwer	Access control, moderation, output access, consent and retention.	FR-12–18; FR-48–56; BR-14/15; privacy requirements.	Interview notes are written; no audio/video recording.
Developer-team discussion, 21–23 Sep 2026	Attendance, recovery, LiveKit intention, free translation engines, secrets and rate limiting.	FR-54–56; FR-60; NFR-04/05/06/15; BR-10/14; CR-01/02/06/08.	Approval pending; CR-08 records current implementation, not a final design choice.

C.1 Requirement family index

Family / IDs	Coverage in this draft	Status
FR-01–03	Authentication and identity	Proposed elaboration; verify exact original wording.
FR-04–11	Meeting create/join/schedule	Proposed elaboration; scheduled invitations open.
FR-12–18	Waiting room, roles, moderation	Proposed elaboration; permission matrix open.
FR-19–27	Audio/video/screen sharing/annotation relation	Proposed elaboration; platform and annotation scope open.
FR-28–33	Captions and translation	Proposed elaboration; language/provider/latency criteria open.
FR-34–40	Chat and reactions	Proposed elaboration; retention/edit/delete rules open.
FR-41–45	Whiteboard and presentation	Proposed elaboration; code language and concurrency behavior open.

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FR-46–47	Host controls	Proposed elaboration; exact control set open.
FR-48–53	Recording and meeting outputs	Proposed elaboration; consent and retention policy open.
FR-54–56	Attendance and report	Exact source wording available in team notes.
FR-57–59	Personalization, apps and support	Exact source wording not available; recover SRC-01.
FR-60	Persist and restore meeting state after restart	Exact source wording available in team notes.
NFR-01–15	Quality requirements	NFR-04/05/06/15 have known descriptions; several other exact wordings are unavailable.
BR-01–15	Business rules	Only BR-10 and BR-14 are described in inspected team notes; recover SRC-01 for others.
CR-01–11	Constraints	CR-01/02/06/08 known; other exact wording must be recovered from SRC-01.